La Scuola d'Italia / Dr. Huson / IB Math Name:
8 May 2026

12.1 Quiz: Complex Numbers

Answer four problems in total. Use the spaces provided for Problems 1 and 2. Answer Problems 3–6 on lined paper.

1. [Maximum mark: 5] AA HL 2023 Nov P1 TZ1 Q07

   It is given that $z = 5 + qi$ satisfies the equation

   $$z^2 + iz = -p + 25i,$$

   where $p, q \in \mathbb{R}$.

   Find the value of p and the value of q.

2. [Maximum mark: 7] AA HL 2021 May P2 TZ1 Q08

Consider the complex numbers

$$z = 2\left(\cos\frac{\pi}{5} + i\sin\frac{\pi}{5}\right)$$

and

$$w = 8\left(\cos\frac{2k\pi}{5} - i\sin\frac{2k\pi}{5}\right),$$

where $k \in \mathbb{Z}^+$.

(a) Find the modulus of zw. [1]

(b) Find the argument of zw in terms of k. [2]

(c) Suppose that $zw \in \mathbb{Z}$. Find the minimum value of k. [3]

(d) For the value of k found in part (c), find the value of zw. [1]

3. [Maximum mark: 22] AA HL 2025 Nov P1 TZ1 Q11

Consider the complex number

$$w = \frac{4i}{1 + i\sqrt{3}}.$$

(a) Show that $w = \sqrt{3} + i$. [2]

(b) Find

(i) $|w|$; [2]

(ii) $\arg w$. [1]

(c) Use de Moivre's theorem to find the two square roots of w. Give your answers in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. [4]

The complex number w can be expressed in the form $(a + bi)^2$, where $a, b \in \mathbb{R}^+$.

(d) (i) Show that $a^2 = \dfrac{\sqrt{3}}{2} + 1$. [6]

(ii) Find the value of b^2. [2]

(e) Hence, express $\tan \frac{\pi}{12}$ in the form $p + q\sqrt{3}$, where $p, q \in \mathbb{Z}$. [5]

4. [Maximum mark: 20] AA HL 2024 Nov P1 TZ0 Q12

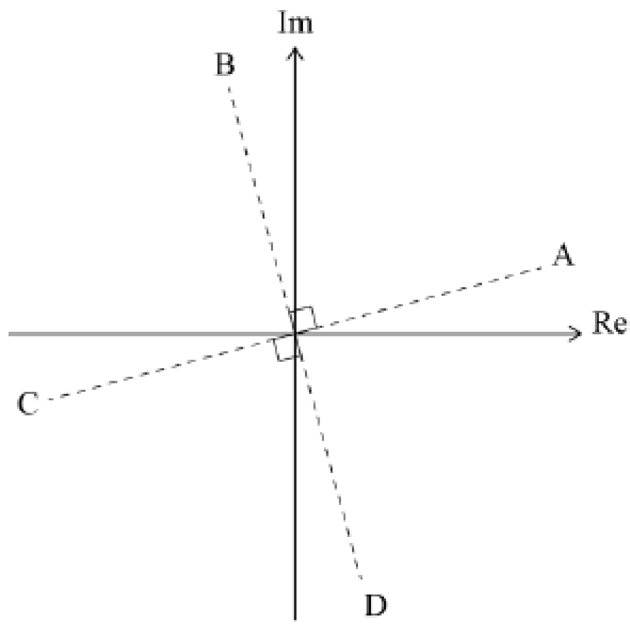
Consider the equation $z^4 = 16i$, where $z \in \mathbb{C}$.

The equation has four roots z_1, z_2, z_3, z_4, where

$$z_j = r\left(\cos\theta_j + i\sin\theta_j\right), \quad r > 0$$

and

$$0 \leq \theta_1 < \theta_2 < \theta_3 < \theta_4 < 2\pi.$$



(a) Find z_1, z_2, z_3 and z_4. [6]

(b) The roots z_1, z_2, z_3 and z_4 form a geometric sequence. Find the common ratio of the sequence, expressing your answer in Cartesian form. [3]

(c) The roots z_1, z_2, z_3 and z_4 are represented by the points A, B, C and D respectively on an Argand diagram. Plot the points A, B, C and D on an Argand diagram. [3]

(d) The equation $v^4 = a + bi$, where $v \in \mathbb{C}$ and $a, b \in \mathbb{R}$, has roots z_1^*, z_2^*, z_3^* and z_4^*. Determine the value of a and the value of b. [3]

The midpoint of [AB] is A′, the midpoint of [BC] is B′, the midpoint of [CD] is C′ and the midpoint of [DA] is D′.

Consider the equation $w^p = 2^q$, where $w \in \mathbb{C}$ and $p, q \in \mathbb{Z}^+$.

Four of the roots of $w^p = 2^q$ are represented by the points A′, B′, C′ and D′.

(e) Find the least possible value of p and the corresponding value of q. [5]

5. [Maximum mark: 22] AA HL 2023 May P1 TZ2 Q11

Consider the complex number $u = -1 + \sqrt{3}i$.

(a) By finding the modulus and argument of u, show that $u = 2e^{i\frac{2\pi}{3}}$. [3]

(b) (i) Find the smallest positive integer n such that u^n is a real number. [3]

(ii) Find the value of u^n when n takes the value found in part (b)(i). [2]

Consider the equation

$$z^3 + 5z^2 + 10z + 12 = 0,$$

where $z \in \mathbb{C}$.

(c) (i) Given that u is a root of $z^3 + 5z^2 + 10z + 12 = 0$, find the other roots. [5]

(ii) By using a suitable transformation from z to w, or otherwise, find the roots of the equation

$$1 + 5w + 10w^2 + 12w^3 = 0,$$

where $w \in \mathbb{C}$. [4]

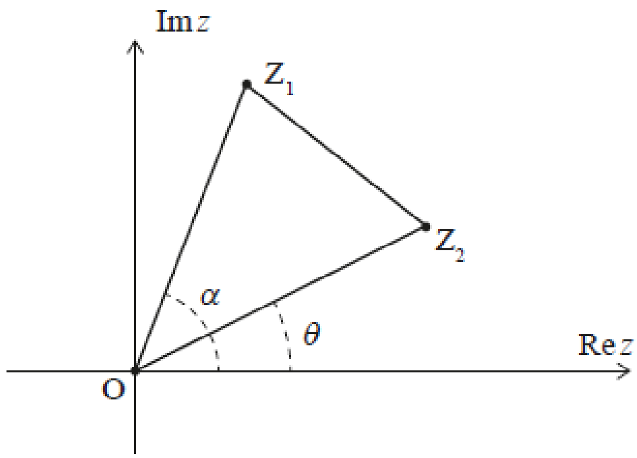
(d) Consider the equation $z^2 = 2z^*$, where $z \in \mathbb{C}$, $z \neq 0$. By expressing z in the form $a + bi$, find the roots of the equation. [5]

6. [Maximum mark: 18] AA HL 2022 May P1 TZ2 Q12

In the following Argand diagram, the points Z_1, O and Z_2 are the vertices of triangle $\mathrm{Z}_1\mathrm{OZ}_2$ described anticlockwise.

The point Z_1 represents the complex number $z_1 = r_1 e^{i\alpha}$, where $r_1 > 0$. The point Z_2 represents the complex number $z_2 = r_2 e^{i\theta}$, where $r_2 > 0$.

Angles α, θ are measured anticlockwise from the positive direction of the real axis such that $0 \leq \alpha, \theta < 2\pi$ and $0 < \alpha - \theta < \pi$.



(a) Show that $z_1 z_2^* = r_1 r_2 e^{i(\alpha - \theta)}$, where z_2^* is the complex conjugate of z_2. [2]

(b) Given that $\mathrm{Re}\,(z_1 z_2^*) = 0$, show that $\mathrm{Z}_1\mathrm{OZ}_2$ is a right-angled triangle. [2]

In parts (c), (d) and (e), consider the case where $\mathrm{Z}_1\mathrm{OZ}_2$ is an equilateral triangle.

(c) (i) Express z_1 in terms of z_2. [2]

(ii) Hence show that $z_1^2 + z_2^2 = z_1 z_2$. [4]

(d) Let z_1 and z_2 be the distinct roots of the equation $z^2 + az + b = 0$ where $z \in \mathbb{C}$ and $a, b \in \mathbb{R}$. Use the result from part (c)(ii) to show that $a^2 - 3b = 0$. [5]

(e) Consider the equation $z^2 + az + 12 = 0$, where $z \in \mathbb{C}$ and $a \in \mathbb{R}$. Given that $0 < \alpha - \theta < \pi$, deduce that only one equilateral triangle $\mathrm{Z}_1\mathrm{OZ}_2$ can be formed from the point O and the roots of this equation. [3]